

tests/dynamic/analyzers/ xcache_hit_analyzer.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active self-validated analyzer (§11.4.107(10)). Signal 4 of 6 for the
dynamic data plane — caching preserved under dynamic mode.

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

Signal 4: **a real Squid cache HIT** (§11.4.69 `storage_read`; design §13 `cache_hit`; research §2). A second request for a cacheable URL must produce a HIT. The **decisive** data-plane fact is the Squid `access.log` carrying a URL-specific `TCP_*HIT` result code (a response header alone is forgeable). It delegates that to the committed, self-tested `evidence.sh:assert_cache_hit`, and — when a captured response-header dump is supplied — additionally requires it to corroborate with X-Cache: HIT (defense in depth: a header that says MISS while the log says HIT is contradictory evidence → FAIL).

Prerequisites

- Source library `tests/dynamic/lib/analyzer_common.sh` + the committed `tests/lib/evidence.sh`.
- `awk`, `grep`, POSIX `sh`.
- A manifest (`key=val`):
 - `access_log=access_hit.log` (resolved relative to the manifest dir)
 - `url=http://cdn.example.com/static/app.css`
 - `headers_file=headers_hit.txt` (optional corroboration)

Usage examples

```
# Analyze a captured cache manifest:  
tests/dynamic/analyzers/xcache_hit_analyzer.sh analyze <manifest-  
file>
```

```
# Self-validate (golden-good PASS + golden-bad FAIL) – the default
  action:
tests/dynamic/analyzers/xcache_hit_analyzer.sh --selftest
```

Edge cases

- **Manifest missing / evidence.sh not found** → FAIL, never a bluff.
- **No TCP_HIT for the URL in the access.log** → FAIL (re-runs the canonical oracle to surface its FAIL line).
- **Declared headers_file missing** → FAIL.
- **access.log shows TCP_HIT but the X-Cache header is not HIT** → FAIL (contradictory evidence).
- **Timing-is-faster is NOT a cache fact** — a result code in the access.log is the data-plane corroboration; a faster second request alone proves nothing.

§11.4.115 RED_MODE polarity

Consuming suites run the §11.4.115 polarity citing this analyzer; RED reproduces an always-MISS regression, GREEN guards the TCP_HIT + X-Cache: HIT contract.

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean (`sh -n + bash -n`, §11.4.67).
- `_xc_manifest_get` parses `key=val` with `awk`; the `log` + optional header file are resolved relative to the manifest dir when not absolute.
- Decisive check first (`assert_cache_hit`), then optional X-Cache header corroboration via `grep -Eiq '^[[:space:]]*X-Cache:.*HIT'`, then `ab_pass_with_evidence` citing the `access.log`.
- Self-test asserts: golden-good PASS; always-MISS, header-contradiction, url-MISS-only → FAIL; missing-manifest negative → FAIL.

Related

- `tests/lib/evidence.sh` (`assert_cache_hit`) — the delegated oracle.
- `tests/dynamic/lib/analyzer_common.sh` — sourced base.
- Fixtures: `tests/dynamic/analyzers/fixtures/xcache_hit/`.
- Constitution §11.4.69 / §11.4.107 / §11.4.115; design §13; research §2.

Last verified

2026-07-01 — self-test PASS (golden-good PASS + all-miss/header-contradiction/ url-miss-only FAIL); sh -n + bash -n parse-clean.
Live cache HIT is exercised in **P10**.